

CASE SUMMARY — Repair Request

QYSEA FIFISH V6 Expert | Saltwater exposure / front thruster failure

Owner / Location	[Your name] — Argentina	Contact	[email] / [phone]
Model	QYSEA FIFISH V6 Expert	Serial no.	[S/N]
Depth version	[M100 / M200]	Tether length	[X] m
Purchased	[dealer, date]	Incident date	2026-07-30

Incident

The ROV was exposed to **saltwater** and was **powered on afterward** (before diagnosis). The only initial sign of moisture was **condensation on the inside of the dome lens**; it is not yet confirmed whether this was benign fog or actual saline ingress (residue test pending).

Observed symptoms (brief bench test in air)

- Only **one thruster** malfunctioned: the **front thruster closest to the dome** — it trembled/shook instead of spinning cleanly.
- The other 5 thrusters ran normally under the same conditions.
- Casing warmed to approx. **40–45 °C** near the affected thruster.
- Per the official manual, thrusters must not run in air for more than 30 s; the test was brief.

Working hypothesis

A **localized failure in that single front thruster** — most likely a **lost phase** (open / high-resistance phase, connector, or ESC channel) due to saline corrosion, consistent with saltwater exposure near the dome. Mechanical obstruction (fouled shaft or corroded bearing) is not yet ruled out.

Current state

- **Battery removed** and stored safely (Li-ion + saltwater = thermal risk).
- **Not re-energized** since the incident. No intervention or disassembly attempted.

Pending non-invasive checks (can be done before shipping)

- Manual spin test of suspect vs. reference thruster (free / rough / locked).
- Phase-to-phase resistance (3 combinations) and insulation phase-to-ground, with multimeter.
- Visual inspection for white (salt) / green-blue (copper) residue at motor leads, pins, penetrator, ESC channel.
- App telemetry: fault code, flagged thruster, battery voltage, leak/temp alarms.

Requested service

- Full diagnosis and disassembly; repair or replacement of the affected front thruster and ESC channel if damaged.
- Cleaning / neutralization of salt corrosion; treatment of any affected electronics.
- **Re-certification of the pressure seal (100–200 m)** with leak/vacuum test before return.
- RMA reference and **shipping guidance from Argentina** (customs, and whether the **Li-ion battery** may ship with the unit under IATA rules).

Attachments: *diagnostico-fifish-v6-expert.md* (detailed diagnostic report). Safety note: Li-ion battery removed from unit; thrusters were not run in air beyond 30 s.